

SCIENCE TECHNOLOGY AND SOCIETY

INFORMATION AGE

The Basics

The Information Age began around the 1970s and continues today. It is also known as the Computer Age, Digital Age, or New Media Age. This era brought about a time period in which people could access information and knowledge easily.

Influences of the past on the Information Age

The Renaissance

- ◇ Influenced the Information Age by creating the idea inventions, while too advanced for the time, the basic idea was used to develop modern inventions.
- ◇ People began to write realistic books and not just religious stories.
- ◇ People started to focus on what they could do for themselves. This way of thinking is called humanism.

The Scientific Revolution

- ◇ Introducing important scientists such as Galileo, Copernicus, and Sir Isaac Newton. Their discoveries paved the way for modern tools, inventions and innovations.

The Industrial Revolution

- ◇ Brought about major changes in agriculture, manufacturing, mining, transportation, and technology. This era had a profound effect on the social, economic, and cultural conditions of the world.

Inventions and Innovations of the Information Age

There were many different inventions during the Information Age, one of which was the computer. The Internet made information accessible with the touch of a button.

It also turned society into homebodies, people can now shop, work, study, and entertain themselves online. While convenient, it has also made many

individuals more dependent and an uneducated society.

I. The Gutenberg Press (1440)

- ◇ Invented by Johannes Gutenberg, a german Goldsmith
- ◇ **printing press** is a device that applies pressure to an inked surface lying on a print medium, such as cloth or paper, to transfer ink
- ◇ Made mass communication fast and easy, permanently reshaping how information is shared in society.

II. World War II and Early Computers

The war created a serious shortage of human computers for military calculations

- ◇ **Allies:** US, Canada, Britain, France, USSR, Australia, etc.
- ◇ **Axis** - germany, japan, italy, hungary, romania, bulgaria
- ◇ **Harvard Mark I** - A general-purpose computer, 50 ft long, capable of doing calculations in seconds that usually took people hours.

III. Alan Turing

An English mathematician hired by the British top-secret Government Code and Cipher School to break the Enigma code

- ◇ **Enigma** - an enciphering machine that the German armed forces used to securely send messages
- ◇ **Bombe** - an electromechanical machine that enabled the British to decipher encrypted messages of the German Enigma machine.
- ◇ **The Universal Turing Machine** - a machine that can solve any problem for simple instructions encoded on a paper tape

IV. The Personal Computer (1970s)

- ◇ Apple co-founders *Steve Wozniak and Steve Jobs* developed the Apple 1, which marked the beginning of personal computing.

◇ **Homebrew Computer Club** - In the 1970s, this early computer hobbyist group gathered regularly to trade parts of computer hardware and talked about how to make computers available to everyone. Many members of this group became high-profile entrepreneurs.

Notable Figures of the Information Age

- ‡ **Tim Berners-Lee** - created the World Wide Web.
- ‡ **Steve Jobs** - created the first effective personal computer called the Apple 1. The Apple 1 was a huge advancement in computer sciences & it carved the path for the computers we have now.
- ‡ **Bill Gates** - a huge force in the Information Age. He founded Microsoft.

Changes of the Information Age

- ‡ The **Internet** is arguably the most prominent innovation of the Information Age. It makes a large amount of the population smarter. It Strengthened industrial countries and boosted global economies through successful online businesses.
- ‡ Helped people become more educated and mature through easier access to information.
- ‡ Reshaped governments with advanced technology and effective militaries.
- ‡ Led to the creation of new laws against hacking, piracy, and identity theft.

Positive Impacts of the Information Age

- ‡ People can learn new languages more easily, and books are widely translated.
- ‡ Jobs have become more flexible, with many tasks now doable from home.
- ‡ The Information Age is also known as the Age of Entrepreneurship.

Negative impacts of the Information Age

- ‡ Some people believe they can live their entire lives online, leading to overdependence on technology.
- ‡ Criminal organizations use hacking to steal confidential government information.
- ‡ Technology affects work ethics, causing distractions and loss of focus.

‡ It has also led to job shortages and made many roles obsolete due to machine automation.

The Age of Social Media: Digital Natives and Immigrants

- ‡ **Digital Natives** - people who are generally born after the 1980s and they are comfortable in the digital age because they grew up using technology.
- ‡ **Digital Immigrants**: people who were born before the 1980s and some are fearful about using technology while others are able to adapt to the age of social media.
- ‡ **Generation with electronic brains** - description given to the generation who witnessed the dawn of the computer age. The people of this generation were the first to be introduced to personal computers (PCs).

BIODIVERSITY

Biodiversity is the variability among living organisms from all sources, including terrestrial, marine, and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species, and of ecosystems.

Biodiversity is the foundation of ecosystem services that support human life, such as clean air, water, fertile soil, food, and a stable climate. It connects the atmosphere, land, and water into one living system through the work of plants, animals, and microbes.

Human actions greatly affect biodiversity, and in turn, influence human well-being. The way people care for the environment determines whether its impacts are positive or harmful.

Types of Biodiversity

- ◇ **Genetic** - The variety of genes within a species that allows adaptation and survival.
- ◇ **Species** - The number and variety of species in a given area.

✧ **Ecosystem** - The variety of habitats and ecosystems within a region.

Types Of Values That Biodiversity Provide

✧ **Intrinsic Value** - It refers to the value of biodiversity inherent to the organism or ecosystem.

✧ **Extrinsic Value** - The value given to biodiversity which grows out of utilitarian or instrumental uses or applications.

✧ **Indirect Use Value** - The value of biodiversity that provides services that include atmospheric and climate regulation, nutrient cycling, pollination, and seed dispersal, among others.

✧ **Non-use Value** - The term referring to less tangible values for the things that we don't use but would feel a loss if they were to disappear.

✧ **Direct Use Value** - The value or biodiversity that provides goods such as food, medicine, building materials, fuel, and fibers for clothing and textile.

✧ **Existence value** - The value of knowing something exists even if you will never use it or see it.

✧ **Bequest Value** - The value of knowing something will be there for future generations

Measuring Biodiversity: Species Richness and Indicators

Biodiversity is hard to measure accurately, but exact numbers aren't always required to understand where it exists, how it changes over time, and its impact on ecosystems and people.

To assess biodiversity trends, scientists look at species abundance, taxonomy (number of species), functional traits (like plant types), and species interactions (e.g., pollination, competition).

However, lack of data makes it difficult to measure biodiversity precisely, and uncertainty remains about its true extent and changes.

Measures Of Biodiversity:

a. Species richness

✧ The number of species in a given area is an important measure of biodiversity, but it must be combined with other metrics for a full picture.

✧ Other measures include plant functional types (grasses, trees) and genetic diversity from soil samples. These help show variability, function, quantity, and distribution, which are key to understanding biodiversity's role in ecosystems.

b. Ecological indicators

✧ use quantitative data to measure biodiversity, ecosystem condition, services, and drivers of change. No single indicator can reflect all aspects of biodiversity, but they play a vital role in monitoring, assessment, and decision-making.

✧ They communicate information clearly to policy-makers, much like economic indicators such as GDP. Widely accepted examples include global temperature and atmospheric CO₂ levels, which reflect human impacts on climate. *Ecological indicators follow the same principles and share similar strengths and limitations.*

Where is biodiversity?

The virtual omnipresence of life on Earth is seldom appreciated because most organisms are small (<5 centimeters); their presence is sparse, ephemeral, or cryptic, or, in the case of microbes, they are invisible to the unaided human eye.

Number of Named Species

- Documenting biodiversity is difficult because many dimensions remain poorly quantified.
- Existing data are biased toward well-known species, megafauna, and temperate regions.

- Major knowledge gaps exist in tropical, marine, freshwater, plant, invertebrate, microbial, and subterranean life.
- The estimated total number of species ranges from 5 to 30 million.
- Only 1.7–2 million species have been formally identified.
- More complete biotic inventories are urgently needed to fill these gaps.

Spatial Patterns of Biodiversity

- ✧ Data on biodiversity distribution are often incomplete but useful patterns and tools exist for estimates.
- ✧ North-temperate regions have better data on species distributions (e.g., birds, mammals, reptiles, plants, butterflies, dragonflies).
- ✧ Biogeographic principles (e.g., latitude, temperature, salinity, water depth) and indicators help supplement limited inventories.
- ✧ Global and sub-global maps of species richness provide valuable biodiversity distribution insights.
- ✧ Most macroscopic organisms have small, clustered ranges—often in isolated or topographically varied regions like islands, mountains, and peninsulas.
- ✧ A large share of terrestrial biodiversity is concentrated in the tropics.
- ✧ Over one third of terrestrial vertebrates have ranges smaller than 1,000 km².
- ✧ Microorganisms show less regional clustering due to large populations, broad ranges, and strong dispersal abilities.

Species Richness in Different Biomes

Biomes and biogeographic realms show the distribution of functional diversity, reflecting ecological patterns in wetlands, forests, grasslands, and estuaries. Ecosystems and ecoregions, as smaller divisions, help estimate functional diversity and how it may change with shifting environmental conditions.

Link between biodiversity and ecosystem services

Biodiversity's role:

- ★ Supports ecosystem functions that provide supporting, provisioning, regulating, and cultural services. These services are essential for human well-being.

Importance of species composition:

- ★ Species composition affects ecosystem services more than species richness.
- ★ Dominant species' traits (e.g., lifespan, size, nutrient use, leaf decomposition, wood density) strongly influence ecosystem processes.
- ★ Conserving or restoring community composition is critical to maintaining services.

Extinctions and ecosystem impacts:

- ★ Local and functional extinctions can disrupt ecosystem functions before global extinction happens.
- ★ Ecosystem-service losses may persist for a long time once functions are pushed beyond limits.

Biotic interactions:

- ★ Changes in interactions (predation, parasitism, competition, facilitation) can lead to large, irreversible impacts.
- ★ Key interactions include pollination, plant–soil relationships, herbivore and seed dispersal links, and habitat modification.

Species removal or introduction:

- ★ Disrupts biotic interactions and ecosystem processes.
- ★ Effects are often complex and difficult to predict.

Real-world example:

- ★ Protecting the Catskill watersheds that supply New York City saved

billions, proving the value of conserving ecosystem services.

Ecological Surprises Caused by Complex Interactions

The loss of key species in marine ecosystems can disrupt ecosystem processes and the services they provide.

Coral reefs, as one of the most species-rich communities on Earth, are especially affected by such disruptions. They maintain a vast storehouse of genetic and biological diversity and provide important ecosystem services, including habitat construction, nurseries and spawning grounds for fish, nutrient cycling, carbon and nitrogen fixing, as well as wave buffering and sediment stabilization.

The total economic value of these reefs and their services is estimated at hundreds of millions of dollars. However, all coral reefs depend on a single crucial biotic interaction: symbiosis with algae. Climate change and events like El Niño oscillations can disrupt this symbiosis, causing significant impacts on coral reefs and the ecosystem services they support.

Why is Biodiversity Important?

Carbon Sequestration, Pollination and Seed Dispersal, Food Source & Maintenance of Food Supply, Nutrient Cycling, Sources of Medicine, Soil and Water Conservation, Habitat, Building Materials, Paper Products, Fuel, Industrial Products, Aesthetic Value

Biodiversity Loss

Ivory Billed Woodpecker

- ★ Declared Extinct: 2021
- ★ global collapse in insect numbers
- ★ 95% drop in population of bluefin tuna

Philippine Crocodile, Philippine Eagle, Tamaraw, Bombon Sardine (Tawilis), Calamian and Philippine spotted deer, Tarsier, Sea Turtles, Balabac mouse deer

(Pilandok), Wild Pig (Baboy Damo), Philippine Forest Turtle, Waling-Waling, Kris Plant, Staghorn Fern, Arabica Coffee, Catmon

Simple Ways to Preserve Biodiversity

- ★ Support local farms
- ★ Save the bees
- ★ Take shorter showers
- ★ Respect local habitats
- ★ Know the source
- ★ We are not above nature.
- ★ We are part of nature.
- ★ Plant local flowers, fruits, & vegetables

CLIMATE CHANGE

✧ **Weather** - short-term changes in the atmosphere. It can change minute-to-minute, hour-to-hour and day-to-day

✧ **Climate** - average weather conditions in a specific area over a long period of time - 30 years or more

✧ **Climate change** - long-term shifts in average weather patterns that affect local, regional, and global climates.

✧ **Global warming** - the long-term heating of Earth's surface observed since the preindustrial period (between 1850 and 1900) due to human activities, primarily fossil fuel burning.

Greenhouse Effect

For centuries, humans have burned fossil fuels to power daily life. This releases greenhouse gases that trap heat in the atmosphere. As a result, the Earth has already warmed by 1 °C, and temperatures may continue to rise, threatening life as we know it.

- ★ The Earth's surface emits infrared radiation after it warms up.
- ★ Some of this infrared radiation passes through the atmosphere, while some is absorbed and re-emitted in all directions by greenhouse gases.

- ★ This process traps heat and warms both the Earth's surface and the lower atmosphere.

Natural Phenomena

In the past 3.5 billion years, the planet's climate has changed because of natural phenomena such as asteroid collisions, volcanic eruptions, solar input, and the shifting of tectonic plates that alter wind and ocean patterns.

Effects of Climate Change

✧ Rise in global temperature

- ★ Earth experienced its warmest years in the last 35 years.
- ★ 2016 had record-high temperatures in 8 of 12 months (except June).

✧ Rise in ocean temperature

- ★ As global temperature rises, ocean temperature increases too.
- ★ This impacts marine life and biodiversity.

✧ El Niño

- ★ Caused by changes in atmospheric circulation in the Pacific.
- ★ Leads to unusually warm ocean surfaces, heat waves, and droughts.

✧ Melting of ice sheets

- ★ Greenland loses 150–250 km³ of ice yearly.
- ★ Antarctica loses about 152 km³ per year (2002–2005).

✧ Glacial retreat

- ★ Glaciers in the Himalayas, Mt. Kilimanjaro, Andes, Rockies, Alaska, and Africa are melting rapidly.
- ★ IPCC warns Kilimanjaro's glaciers may disappear by 2050.

✧ Decreasing snow cover

- ★ Spring snow cover in the Northern Hemisphere has greatly decreased in 50 years.
- ★ This threatens biodiversity and Arctic wildlife.

✧ Rise in sea level

- ★ Melting ice and snow raise sea levels by about 3 mm per year.
- ★ Causes flooding, erosion, soil contamination, and habitat loss.

✧ Extreme events

- ★ Temperatures have been reaching higher levels in warmer climates and lower levels in colder climates.
- ★ The number of episodes of intense rainfall and floods is increasing, causing massive destruction to communities.

Three Ways that Climate Change is already affecting people's lives

1. Health

- ✧ Climate change impacts population health by affecting clean air, water, food, and shelter security
- ✧ The same dirty fossil fuel emissions that contribute to the greenhouse effect can lead to respiratory diseases - such as asthma
- ✧ Air pollution kills an estimated 7 million people worldwide every year.
- ✧ By trapping heat into our planet, carbon emissions damage the human body and mind in other ways.

2. Home

- ✧ Rising sea levels (7–8 inches since 1900, 3 inches since 1993) are forcing coastal communities to relocate.
- ✧ Flooding is increasing in the U.S. East Coast flood days more than doubled (2005–2015).
- ✧ In places like Miami, even inland residents may be displaced.
- ✧ Lower-income groups, people of color, and immigrants are at higher risk due to "*climate gentrification*", where wealthy residents move away from the coast, driving up property prices.

3. Food

- ✧ CO₂ buildup is changing the nutritional value of fruits and vegetables.
- ✧ Extra CO₂ speeds up photosynthesis, causing plants to have more sugar but less calcium, protein, zinc, and vitamins.
- ✧ Harvard researchers warn that by mid-century, 175 million people could face zinc deficiency and 122 million protein deficiency if emissions aren't reduced.

Social & Economic Impact Of Climate Change

- ★ The cost of adapting coastal areas to rising sea levels
- ★ Loss of the capacity to work due to heat
- ★ More wars to gain access to limited resources
- ★ Fresh water will be in short supply in some areas
- ★ Diseases will spread due to higher temperatures
- ★ Extreme meteorological phenomena will cause widespread poverty
- ★ Prices of basic foodstuffs and consumer goods will rise
- ★ Shrinking productivity of harvests
- ★ Relocation of whole towns

Vulnerable Groups

Communities of color - Often live in high-risk areas with multiple pollutant exposures. Adaptation plans and better healthcare access can reduce social inequities.

Children - have a higher risk of heat stroke and illness than adults.

Older Adults - vulnerable to extreme events that cause power outages or require evacuation

Low-income communities - at risk of physical and mental illnesses during flooding and in crowded shelter conditions. Comprehensive disaster management can improve resiliency for people with limited resources.

ENVIRONMENTAL AWARENESS

Environmentalism - an ideology that evokes the necessity and responsibility of humans to respect, protect, and preserve the natural world from its anthropogenic (caused by humans) afflictions.

Environmental awareness - by teaching others about the fragility of the environment helps address environmental problems.

Several Cause-And-Effect Problems That Harm Our Environment:

I. Oil Drilling

- ★ Fossil fuel dependence causes oil spills, marine poisoning, and global warming.
- ★ Offshore drilling damages oceans; land drilling harms the soil and contributes to ocean acidification.

II. Deforestation

- ★ Forests are cleared for industrial benefit, such as large-scale farming, oil mining, and the production of paper goods.
- ★ Causes wildlife and biodiversity extinction because the loss of habitat threatens many species' existence

III. Production of Plastic Goods

- ★ In 2010, 31 million tons were created, spreading across land and water — including the Great Pacific Garbage Patch. Plastic production also depends on fossil fuel combustion. That same year, 191 million barrels of petroleum gases (LPG) and natural gas liquids (NGL) were used in the U.S. to make plastic goods.

